

The energy of the electron is a function of the wave vector and is written as  $\mathcal{E} = \hbar^2 k^2 / 2m^*$  for an electron with isotropic effective mass  $m^*$ , which leads to the relation  $\hbar k_x = m^* v_x$ . Therefore, we obtain

$$f_1 = -\frac{\tau}{\hbar} F_x \frac{\partial f}{\partial \mathcal{E}} \frac{\partial \mathcal{E}}{\partial k_x} = -\tau v_x F_x \frac{\partial f}{\partial \mathcal{E}}. \quad (6.92)$$

When we use the relations  $f = f_0 + f_1$  and  $f_0 \gg f_1$  in the above equation we may approximate it as

$$f_1 = -\tau v_x F_x \frac{\partial f_0}{\partial \mathcal{E}}. \quad (6.93)$$

When we assume elastic scattering and that the relaxation time  $\tau$  is a function of the energy  $\mathcal{E}$ , the magnitude of the relaxation time  $\tau$  is not changed after the scattering event and the relaxation time  $\tau(\mathbf{k})$  is expressed as follows by inserting (6.93) into (6.86):

$$\begin{aligned} \frac{1}{\tau(\mathbf{k})} &= \frac{V}{(2\pi)^3} \int d^3 \mathbf{k}' P(\mathbf{k}, \mathbf{k}') \left(1 - \frac{k'_x}{k_x}\right) \\ &= \frac{V}{(2\pi)^3} \int d^3 \mathbf{k}' P(\mathbf{k}, \mathbf{k}') (1 - \cos \theta), \end{aligned} \quad (6.94)$$

where  $k_x$  and  $k'_x$  are the components of the electron wave vectors  $\mathbf{k}$  and  $\mathbf{k}'$  along the direction of the external force ( $x$  component) before and after scattering, respectively, and  $\theta$  is the angle between  $\mathbf{k}$  and  $\mathbf{k}'$ .

In the case of a degenerate semiconductor, the principle of detailed balance is obtained from (6.80) as

$$P(\mathbf{k}', \mathbf{k}) f_0(\mathbf{k}') [1 - f_0(\mathbf{k})] = P(\mathbf{k}, \mathbf{k}') f_0(\mathbf{k}) [1 - f_0(\mathbf{k}')] . \quad (6.95)$$

Using this relation, the relaxation time of a degenerate semiconductor is given by

$$\begin{aligned} \frac{1}{\tau(\mathbf{k})} &= \frac{1}{1 - f_0(\mathbf{k})} \sum_{\mathbf{k}'} P(\mathbf{k}, \mathbf{k}') [1 - f_0(\mathbf{k}')] \left(1 - \frac{k'_x}{k_x}\right) \\ &= \frac{1}{1 - f_0(\mathbf{k})} \frac{V}{(2\pi)^3} \int d\mathbf{k}' P(\mathbf{k}, \mathbf{k}') [1 - f_0(\mathbf{k}')] \left(1 - \frac{k'_x}{k_x}\right). \end{aligned} \quad (6.96)$$

The above equation reduces to (6.94) for the non-degenerate case when we let  $f_0(\mathbf{k}) \ll 1$  and  $f_0(\mathbf{k}') \ll 1$ . It may be pointed out that when the final state is occupied by an electron,  $1 - f_0(\mathbf{k}') = 0$  and the transition (scattering) is not allowed.

### 6.2.2 Mobility and Electrical Conductivity

When the relaxation time is given by a function of the electron energy or electron wave vector, the electron mobility and electrical conductivity (referred

to as mobility and conductivity hereafter, respectively, unless otherwise specified) are calculated as follows. In the presence of an electric field applied in the  $x$  direction, we may write  $F_x = -eE_x$ ; (6.84) and (6.93) then give the following relation:

$$f(\mathbf{k}) = f_0(\mathbf{k}) + eE_x \tau v_x \frac{\partial f_0}{\partial \mathcal{E}}. \quad (6.97)$$

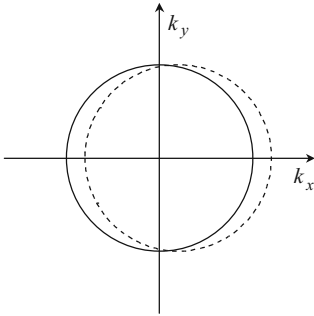
Therefore the current density in the  $x$  direction is given by

$$\begin{aligned} J_x &= \frac{2}{(2\pi)^3} \int (-e) v_x f(\mathbf{k}) d^3\mathbf{k} \\ &= -\frac{e}{4\pi^3} \int v_x f_0(\mathbf{k}) d^3\mathbf{k} - \frac{e^2 E_x}{4\pi^3} \int \tau v_x^2 \frac{\partial f_0}{\partial \mathcal{E}} d^3\mathbf{k}, \end{aligned} \quad (6.98)$$

where the factor 2 arises from the spin degeneracy and  $v_x = \hbar k_x / m^*$ . The function  $f_0(\mathbf{k})$  is given by the Fermi (or Fermi–Dirac) distribution function or the Boltzmann distribution function, which is a function of the electron energy  $\mathcal{E}(\mathbf{k})$ . Since  $\mathcal{E}$  is an even function of  $\mathbf{k}$ ,  $v_x f_0(\mathbf{k})$  is an odd function of  $v_x$ . Since integration with respect to  $dk_x$  ranges from  $-\infty$  to  $+\infty$ , the first term on the right hand side becomes zero, and only the second term remains, resulting in

$$J_x = -\frac{e^2 E_x}{4\pi^3} \int \tau v_x^2 \frac{\partial f}{\partial \mathcal{E}} d^3\mathbf{k}. \quad (6.99)$$

This result is interpreted with the help of Fig. 6.6. In the absence of an electric field, the distribution function of the thermal equilibrium state is isotropic in  $\mathbf{k}$  space and thus the current in any direction has the same magnitude, resulting in mutual cancellation and hence in zero current. In the presence of an applied electric field, the distribution function is given by  $f = f_0 + f_1$  and is displaced by  $f_1$  in the electric field direction, resulting in a current proportional to the displacement along the electric field direction. Using the electron density  $n$  given by



**Fig. 6.6.** Change in the distribution function under an applied electric field. The distribution function in  $\mathbf{k}$  space is slightly displaced along the field direction

$$n = \frac{2}{(2\pi)^3} \int f_0 d^3 \mathbf{k}, \quad (6.100)$$

(6.99) is rewritten as

$$J_x = -e^2 n E_x \frac{\int \tau v_x^2 \frac{df_0}{d\mathcal{E}} d^3 \mathbf{k}}{\int f_0 d^3 \mathbf{k}} = \frac{e^2 n E_x}{k_B T} \frac{\int \tau v_x^2 f_0 (1 - f_0) d^3 \mathbf{k}}{\int f_0 d^3 \mathbf{k}}, \quad (6.101)$$

where the final result is obtained by using the following relations:

$$f_0(\mathcal{E}) = \frac{1}{e^{(\mathcal{E} - \mathcal{E}_F)/k_B T} + 1} \quad (6.102)$$

$$\frac{\partial f_0}{\partial \mathcal{E}} = -\frac{1}{k_B T} f_0 (1 - f_0). \quad (6.103)$$

Since the term  $\tau f_0 (1 - f_0)$  in the integral of (6.101) is a function of the energy  $\mathcal{E}$ , we express it as  $\phi(\mathcal{E})$ . Noting  $d^3 \mathbf{k} = dk_x dk_y dk_z$ , we have

$$\int v_x^2 \phi(\mathcal{E}) d^3 \mathbf{k} = \int v_y^2 \phi(\mathcal{E}) d^3 \mathbf{k} = \int v_z^2 \phi(\mathcal{E}) d^3 \mathbf{k} = \frac{1}{3} \int v^2 \phi(\mathcal{E}) d^3 \mathbf{k}. \quad (6.104)$$

The integral with respect to  $d^3 \mathbf{k}$  is changed into an integral with respect to  $\mathcal{E}$  by using the relation  $\mathcal{E} = \hbar^2 k^2 / 2m^*$ . We then obtain

$$d^3 \mathbf{k} = 4\pi k^2 dk = \frac{8\pi m^{*3/2}}{\sqrt{2}\hbar^3} \mathcal{E}^{1/2} d\mathcal{E} \equiv A \mathcal{E}^{1/2} d\mathcal{E}. \quad (6.105)$$

In the above equation the effective mass  $m^*$  is the density-of-states mass for the isotropic (scalar) effective mass and is often notified by  $m_d^*$ . In semiconductors such as Ge and Si, we know that the conduction bands consist of a many-valley structure and their constant energy surface is expressed by an ellipsoid with transverse effective mass  $m_t$  and longitudinal effective mass  $m_l$ . In such a case the density-of-states mass  $m_d^*$  is given by  $(m_t^2 m_l)^{1/3} \equiv m_d^*$ . Inserting this relation, (6.104) and (6.105) into (6.101), we obtain

$$J_x = \frac{2e^2 n E_x}{3k_B T m^*} \frac{\int_0^\infty \tau \mathcal{E}^{3/2} f_0 (1 - f_0) d\mathcal{E}}{\int_0^\infty \mathcal{E}^{1/2} f_0 d\mathcal{E}}. \quad (6.106)$$

Although the upper limit of the integral should be the upper edge of the band, it is replaced by infinity, taking account of the exponential decay of the distribution function in the higher energy region. As stated above, the density-of-states mass appears in the denominator and numerator and is cancelled, which is evident from (6.101). The effective mass  $m^*$  in (6.106) results from the term  $v_x = \hbar k_x / m^*$ , assuming an isotropic effective mass. The effective mass, usually defined by  $m_c^*$ , is called the *conductivity effective mass* because

it comes from the term for the velocity component. In the case of many-valley semiconductors such as Ge and Si, however, the energy is expressed by an ellipsoid, and the conductivity effective mass is given by  $1/m_c^* \equiv (1/3)(2/m_t + 1/m_l)$ . Next, we carry out the calculation of (6.106) for the following two cases separately.

(a) *Metal and Degenerate Semiconductor.* In this case the Fermi energy  $\mathcal{E}_F$  is located in the conduction band, and the term  $-df_0/d\mathcal{E} = f_0(1 - f_0)/k_B T$  has a large value near the Fermi energy only. Therefore, the term  $-df_0/d\mathcal{E}$  may be approximated by the Dirac  $\delta$ -function. In addition, the following relations,

$$f_0 \cong 1 \quad (\mathcal{E} \leq \mathcal{E}_F),$$

$$f_0 \cong 0 \quad (\mathcal{E} > \mathcal{E}_F),$$

hold for the Fermi distribution function, and therefore we obtain

$$\begin{aligned} & \frac{1}{k_B T} \int_0^\infty \tau(\mathcal{E}) \mathcal{E}^{3/2} f_0 (1 - f_0) d\mathcal{E} \\ &= - \int_0^\infty \tau(\mathcal{E}) \mathcal{E}^{3/2} \frac{df_0}{d\mathcal{E}} d\mathcal{E} \cong \tau(\mathcal{E}_F) \mathcal{E}_F^{3/2} \end{aligned} \quad (6.107)$$

$$\int_0^\infty \mathcal{E}^{1/2} f_0 d\mathcal{E} = \int_0^{\mathcal{E}_F} \mathcal{E}^{1/2} d\mathcal{E} = \frac{2}{3} \mathcal{E}_F^{3/2}. \quad (6.108)$$

Inserting these relations into (6.106), we obtain

$$J_x = \frac{n e^2 \tau(\mathcal{E}_F)}{m^*} E_x, \quad (6.109)$$

where  $\tau(\mathcal{E}_F)$  is the relaxation time of an electron at  $\mathcal{E} = \mathcal{E}_F$  (or at the Fermi surface). The relation between the Fermi energy  $\mathcal{E}_F$  and the electron density  $n$  is obtained from (6.100) and (6.107), and is given by

$$\mathcal{E}_F = \frac{\hbar^2}{2m^*} (3\pi^2 n)^{2/3}. \quad (6.110)$$

(b) *Non-degenerate semiconductor.* For non-degenerate semiconductors, we have  $f_0 \ll 1$  and  $1 - f_0 \cong 1$  and we may approximate as

$$f_0 = \exp\left(\frac{\mathcal{E}_F}{k_B T}\right) \exp\left(-\frac{\mathcal{E}}{k_B T}\right) \equiv A_F \exp\left(-\frac{\mathcal{E}}{k_B T}\right), \quad (6.111)$$

and (6.106) reduces to

$$J_x = \frac{2e^2 n E_x}{3k_B T m^*} \frac{\int_0^\infty \tau \mathcal{E}^{3/2} f_0 d\mathcal{E}}{\int_0^\infty \mathcal{E}^{1/2} f_0 d\mathcal{E}}. \quad (6.112)$$

Integration by parts gives rise to

$$\int_0^\infty \mathcal{E}^{3/2} f_0 d\mathcal{E} = \frac{3}{2} k_B T \int_0^\infty \mathcal{E}^{1/2} f_0 d\mathcal{E}. \quad (6.113)$$

Using this relation we obtain following relation:

$$J_x = \frac{e^2 n E_x}{m^*} \frac{\int_0^\infty \tau \mathcal{E}^{3/2} f_0 d\mathcal{E}}{\int_0^\infty \mathcal{E}^{3/2} f_0 d\mathcal{E}}. \quad (6.114)$$

When we rewrite (6.114) as

$$J_x = \frac{ne^2 \langle \tau \rangle}{m^*} E_x, \quad (6.115)$$

the relaxation time  $\langle \tau \rangle$  is given by

$$\langle \tau \rangle = \frac{\int_0^\infty \tau \mathcal{E}^{3/2} f_0 d\mathcal{E}}{\int_0^\infty \mathcal{E}^{3/2} f_0 d\mathcal{E}} \quad (6.116)$$

and  $\langle \tau \rangle$  represents the average of  $\tau(\mathcal{E})$ .

When we define the average velocity of the electron by  $\langle v_x \rangle$  in the presence of an electric field in the  $x$  direction, it is also calculated from (6.116). The average velocity is often referred to as the *drift velocity*. This is because the electron contributes to the current via *drift motion* along the electric field direction, suffering from collisions. The average time between the scattering events (collisions) or the scattering rate (collision rate) is given by the average value of the relaxation time  $\langle \tau \rangle$ , where the scattering rate is not equivalent to the relaxation time in the strict sense. The difference will be discussed later. Using the electron density  $n$ , the current density is written as

$$J_x = n(-e) \langle v_x \rangle, \quad (6.117)$$

where the drift velocity  $\langle v_x \rangle$  is given by (6.115) as

$$\langle v_x \rangle = -\frac{e \langle \tau \rangle}{m^*} E_x \equiv -\mu E_x. \quad (6.118)$$

Here  $\mu$  is the average electron velocity (drift velocity) under a unit electric field and gives a measure of the mobility; it is called the *electron mobility*. This mobility is often referred to as the drift mobility to distinguish it from the Hall mobility defined in Chap. 7. Using  $\langle \tau \rangle$  the electron mobility is expressed as

$$\mu = \frac{e \langle \tau \rangle}{m^*}, \quad (6.119)$$

and when we define the *conductivity* (electrical conductivity)  $\sigma$  by

$$J_x = \sigma E_x , \quad (6.120)$$

the conductivity is given by

$$\sigma = \frac{ne^2 \langle \tau \rangle}{m^*} = ne\mu . \quad (6.121)$$

These relations hold for degenerate semiconductors by replacing  $\langle \tau(\mathcal{E}) \rangle$  with  $\tau(\mathcal{E}_F)$ .

As a simple example we consider the case of the relaxation time  $\tau$  given by a function of energy  $\mathcal{E}$  such as

$$\tau = a\mathcal{E}^{-s} . \quad (6.122)$$

Then from (6.116) we obtain

$$\langle \tau \rangle = \frac{a \int_0^\infty \mathcal{E}^{3/2-s} \exp(-\mathcal{E}/k_B T) d\mathcal{E}}{\int_0^\infty \mathcal{E}^{3/2} \exp(-\mathcal{E}/k_B T) d\mathcal{E}} = a(k_B T)^{-s} \frac{\Gamma(\frac{5}{2} - s)}{\Gamma(\frac{5}{2})} , \quad (6.123)$$

where  $\Gamma(s)$  is the Gamma function and has properties such as

$$\Gamma(s) = \int_0^\infty x^{s-1} e^{-x} dx , \quad (6.124)$$

$$\Gamma(s+1) = s\Gamma(s), \quad \Gamma(1) = 1, \quad \Gamma(\tfrac{1}{2}) = \sqrt{\pi} . \quad (6.125)$$

## 6.3 Scattering Probability and Transition Matrix Element

### 6.3.1 Transition Matrix Element

The transition probability is calculated with the help of quantum mechanics as follows. Let us define the unperturbed Hamiltonian by  $H_0$ , the time-dependent perturbing Hamiltonian by  $H_1$  and the total Hamiltonian by  $H$ . The eigenstates of the unperturbed Hamiltonian are assumed to be

$$H = H_0 + H_1 , \quad (6.126)$$

$$H_0 |\mathbf{k}\rangle = \mathcal{E}(\mathbf{k}) |\mathbf{k}\rangle . \quad (6.127)$$

The scattering rate for electron scattered from an initial state  $|\mathbf{k}\rangle$  to a final state  $|\mathbf{k}'\rangle$  is then given by the following relation according to quantum mechanics:

$$P(\mathbf{k}, \mathbf{k}') = \frac{2\pi}{\hbar} |\langle \mathbf{k}' | H_1 | \mathbf{k} \rangle|^2 \delta[\mathcal{E}_{\mathbf{k}'} - \mathcal{E}_{\mathbf{k}}] , \quad (6.128)$$

where  $\mathcal{E}_{\mathbf{k}}$  and  $\mathcal{E}_{\mathbf{k}'}$  are the energies of the initial and final states including the perturbation state and ensures the  $\delta$ -function indicate energy conservation. This relation can also be written as follows. The scattering probability  $w$  is calculated by taking all the possible final states into account and is written as

$$\begin{aligned} w &= \frac{2\pi}{\hbar} \sum_{\mathbf{k}'} |\langle \mathbf{k}' | H_1 | \mathbf{k} \rangle|^2 \delta[\mathcal{E}_{\mathbf{k}'} - \mathcal{E}_{\mathbf{k}}] \\ &= \frac{2\pi}{\hbar} \frac{L^3}{(2\pi)^3} \int d^3\mathbf{k}' |\langle \mathbf{k}' | H_1 | \mathbf{k} \rangle|^2 \delta[\mathcal{E}_{\mathbf{k}'} - \mathcal{E}_{\mathbf{k}}] . \end{aligned} \quad (6.129)$$

Next we consider electron scattering by phonons. Using (6.58) we obtain

$$H = H_e + H_1 + H_{e1} , \quad (6.130)$$

$$H_e |\mathbf{k}\rangle = \mathcal{E}(\mathbf{k}) |\mathbf{k}\rangle , \quad (6.131)$$

$$H_1 |n_{\mathbf{q}}\rangle = \hbar\omega_{\mathbf{q}} (n_{\mathbf{q}} + \frac{1}{2}) |n_{\mathbf{q}}\rangle , \quad (6.132)$$

where  $H_{e1}$  is the electron-phonon interaction Hamiltonian. Then the scattering probability for electrons from  $|\mathbf{k}\rangle$  to  $|\mathbf{k}'\rangle$  states is written as

$$P(\mathbf{k}, \mathbf{k}') = \frac{2\pi}{\hbar} |\langle \mathbf{k}', \mathbf{q}' | H_{e1} | \mathbf{k}, \mathbf{q} \rangle|^2 \delta[\mathcal{E}(\mathbf{k}') - \mathcal{E}(\mathbf{k}) \mp \hbar\omega_{\mathbf{q}}] , \quad (6.133)$$

where  $|n_{\mathbf{q}}\rangle$  is written as  $|\mathbf{q}\rangle$  for simplicity. It should be noted that the phonon energy  $\hbar\omega_{\mathbf{q}}$  involved with phonon emission or absorption is included in the  $\delta$ -function.

From the Boltzmann transport equation and the principle of detailed balance, or from (6.94), the relaxation time is given by

$$\begin{aligned} \frac{1}{\tau(\mathbf{k})} &= \sum_{\mathbf{k}'} P(\mathbf{k}, \mathbf{k}') \left( 1 - \frac{k'_x}{k_x} \right) = \sum_{\mathbf{k}'} P(\mathbf{k}, \mathbf{k}') (1 - \cos \theta) \\ &= \frac{L^3}{(2\pi)^3} \int d^3\mathbf{k}' P(\mathbf{k}, \mathbf{k}') (1 - \cos \theta) , \end{aligned} \quad (6.134)$$

where the volume  $V$  is replaced by  $L^3$  in the final relation. As defined previously,  $P(\mathbf{k}, \mathbf{k}')$  is the transition rate of electrons from state  $\mathbf{k}$  to state  $\mathbf{k}'$ , and  $\theta$  is the angle between  $\mathbf{k}$  and  $\mathbf{k}'$ . The transition rate  $P(\mathbf{k}, \mathbf{k}')$  is

$$P(\mathbf{k}, \mathbf{k}') = \frac{2\pi}{\hbar} |\langle \mathbf{k}', \mathbf{q}' | V_s | \mathbf{k}, \mathbf{q} \rangle|^2 \delta[\mathcal{E}_{\mathbf{k}', \mathbf{q}'} - \mathcal{E}_{\mathbf{k}, \mathbf{q}}] , \quad (6.135)$$

where  $V_s$  is the scattering potential and is equivalent to the perturbation Hamiltonian  $H_1$  defined above, and  $|\mathbf{k}, \mathbf{q}\rangle$  is given by the product of the electron wave function  $|\mathbf{k}\rangle$  with wave vector  $\mathbf{k}$  and the wave function for the scattering center, which is a phonon  $|\mathbf{q}\rangle$  with wave vector  $\mathbf{q}$ . Therefore, the relaxation time is given by

$$\begin{aligned}
\frac{1}{\tau(\mathbf{k})} &= \frac{2\pi}{\hbar} \sum_{\mathbf{k}'} |\langle \mathbf{k}', \mathbf{q}' | V | \mathbf{k}, \mathbf{q} \rangle|^2 (1 - \cos \theta) \delta[\mathcal{E}_{\mathbf{k}', \mathbf{q}'} - \mathcal{E}_{\mathbf{k}, \mathbf{q}}] \\
&= \frac{L^3}{(2\pi)^2 \hbar} \int d^3 \mathbf{k}' |\langle \mathbf{k}', \mathbf{q}' | V | \mathbf{k}, \mathbf{q} \rangle|^2 (1 - \cos \theta) \delta[\mathcal{E}_{\mathbf{k}', \mathbf{q}'} - \mathcal{E}_{\mathbf{k}, \mathbf{q}}] .
\end{aligned} \tag{6.136}$$

In the following we use the relation

$$\sum_{\mathbf{k}'} = \frac{L^d}{(2\pi)^d} \int d^d \mathbf{k}' . \tag{6.137}$$

Here we will describe a very convenient method to derive the matrix element for scattering, where we use the Fourier transform shown in Appendix A.3. This is done by using cyclic boundary conditions as follows. Let us consider a semiconductor of length  $L$  and dimension  $d$ . The Fourier transform is written as

$$V_s(\mathbf{q}) = \frac{1}{L^d} \int d^d \mathbf{r} V_s(\mathbf{r}) e^{-i\mathbf{q} \cdot \mathbf{r}} , \tag{6.138a}$$

$$V_s(\mathbf{r}) = \sum_{\mathbf{q}} V_s(\mathbf{q}) e^{i\mathbf{q} \cdot \mathbf{r}} . \tag{6.138b}$$

Next we use the relation

$$\frac{1}{L^d} \int d^d \mathbf{r} \exp[i(\mathbf{q} - \mathbf{q}') \cdot \mathbf{r}] = \delta_{\mathbf{q}, \mathbf{q}'}, \tag{6.139}$$

of which proof is given in A.2 of Appendix A. Then the matrix element is expressed as

$$\begin{aligned}
\langle \mathbf{k}', \mathbf{q}' | V_s(\mathbf{r}) | \mathbf{k}, \mathbf{q} \rangle &= \sum_{\mathbf{q}''} \langle \mathbf{q}' | V_s(\mathbf{q}'') | \mathbf{q} \rangle \langle \mathbf{k}' | e^{i\mathbf{q}'' \cdot \mathbf{r}} | \mathbf{k} \rangle \\
&= \sum_{\mathbf{q}''} \langle \mathbf{q}' | V_s(\mathbf{q}'') | \mathbf{q} \rangle \delta_{\mathbf{k}' - \mathbf{k}, \mathbf{q}''} \\
&= \langle \mathbf{q}' | V_s(\mathbf{k}' - \mathbf{k}) | \mathbf{q} \rangle .
\end{aligned} \tag{6.140}$$

The above equation is derived by using the electron wave function

$$|\mathbf{k}\rangle = \sqrt{\frac{1}{L^3}} \exp(i\mathbf{k} \cdot \mathbf{r}) . \tag{6.141}$$

If we use the Bloch function

$$|\mathbf{k}\rangle = \sqrt{\frac{1}{L^3}} u_{\mathbf{k}}(\mathbf{r}) \exp(i\mathbf{k} \cdot \mathbf{r}) \tag{6.142}$$

for the electron, then we obtain following result. When we use the cyclic boundary conditions for the Bloch function, the integral is decomposed into the integral in a unit cell and the summation over the unit cells.



$$\begin{aligned}
& \langle \mathbf{k}' | \exp(i\mathbf{q}'' \cdot \mathbf{r}) | \mathbf{k} \rangle \\
&= \frac{1}{N\Omega} \int d^3\mathbf{r} u_{\mathbf{k}'}^*(\mathbf{r}) u_{\mathbf{k}}(\mathbf{r}) \exp[i(\mathbf{k} - \mathbf{k}' + \mathbf{q}'') \cdot \mathbf{r}] \\
&= \frac{1}{N} \sum_j \exp[i(\mathbf{k} - \mathbf{k}' + \mathbf{q}'') \cdot \mathbf{R}_j] \\
&\quad \times \frac{1}{\Omega} \int_{\Omega} u_{\mathbf{k}'}^*(\mathbf{r}) u_{\mathbf{k}}(\mathbf{r}) \exp[i(\mathbf{k} - \mathbf{k}' + \mathbf{q}'') \cdot \mathbf{r}] d^3\mathbf{r} \\
&= \delta_{\mathbf{k}' - \mathbf{k}, \mathbf{q}''} \frac{1}{\Omega} \int_{\Omega} u_{\mathbf{k}'}^*(\mathbf{r}) u_{\mathbf{k}}(\mathbf{r}) d^3\mathbf{r} \\
&= I(\mathbf{k}, \mathbf{k}') \delta_{\mathbf{k}' - \mathbf{k}, \mathbf{q}''} .
\end{aligned} \tag{6.143}$$

In the above equations the summation with respect to  $j$  is non zero only when  $\mathbf{k} - \mathbf{k}' + \mathbf{q}'' + \mathbf{G} = 0$  ( $\mathbf{G}$ : reciprocal lattice vector), and thus the Kronecker  $\delta$ -function should be written as  $\delta_{\mathbf{k}' - \mathbf{k}, \mathbf{q}'' + \mathbf{G}}$ . Here, however, we assume that the *Umklapp process* of  $\mathbf{G} \neq 0$  (Umklapp process) does not play a part in the scattering process and that the *normal process* of  $\mathbf{G} = 0$  takes part in the scattering process. In addition, we have

$$I(\mathbf{k}, \mathbf{k}') = \frac{1}{\Omega} \int_{\Omega} u_{\mathbf{k}'}^*(\mathbf{r}) u_{\mathbf{k}}(\mathbf{r}) d^3\mathbf{r} \simeq 1 , \tag{6.144}$$

where  $\mathbf{R}_j$  is a lattice vector,  $N$  is the number of unit cells and  $\Omega$  is the volume of a unit cell.

In the following, we will calculate the matrix elements for several scattering processes which play an important role in transport in semiconductors. It should be noted that notations such as  $|\langle \mathbf{k}' | H_1 | \mathbf{k} \rangle|$ ,  $|V(\mathbf{k}' - \mathbf{k})|$  and  $|M(\mathbf{k}, \mathbf{k}')|$  are used in this book and that all of them represent the matrix elements for the scattering centers.

### 6.3.2 Deformation Potential Scattering (Acoustic Phonon Scattering)

The deformation potential theory has been proposed by Bardeen and Shockley, which is based on the energy change of an electron due to lattice deformation. The energy change is related to the volume change of a crystal,  $\Delta(\mathbf{r}) = \delta V/V$  ( $V$ : volume), and the electron-phonon interaction Hamiltonian  $H_{\text{el}}$  is defined by

$$H_{\text{el}} = D_{\text{ac}} \frac{\delta V}{V} = D_{\text{ac}} \text{div } \mathbf{u}(\mathbf{r}) , \tag{6.145}$$

where  $D_{\text{ac}}$  is called the *deformation potential* for electron scattering by acoustic phonons. The displacement vector of an acoustic phonon  $\mathbf{u}(\mathbf{r})$  has been derived in Sect. 6.1 and is given by

$$\mathbf{u}(\mathbf{r}) = \sum_{\mathbf{q}} \sqrt{\frac{\hbar}{2MN\omega_{\mathbf{q}}}} \mathbf{e}_{\mathbf{q}} [a_{\mathbf{q}} e^{i\mathbf{q}\cdot\mathbf{r}} + a_{\mathbf{q}}^{\dagger} e^{-i\mathbf{q}\cdot\mathbf{r}}], \quad (6.146)$$

where  $\mathbf{e}_{\mathbf{q}}$  is the unit vector along the displacement direction. This result leads to the following relation:

$$H_{\text{el}} = D_{\text{ac}} \sum_{\mathbf{q}} \sqrt{\frac{\hbar}{2MN\omega_{\mathbf{q}}}} (i\mathbf{e}_{\mathbf{q}} \cdot \mathbf{q}) [a_{\mathbf{q}} e^{i\mathbf{q}\cdot\mathbf{r}} - a_{\mathbf{q}}^{\dagger} e^{-i\mathbf{q}\cdot\mathbf{r}}]. \quad (6.147)$$

Next, we calculate the matrix element for this interaction Hamiltonian. Here we replace  $|\mathbf{q}\rangle$  by  $|n_{\mathbf{q}}\rangle$  and use the following relations:

$$a_{\mathbf{q}} |n_{\mathbf{q}}\rangle = \sqrt{n_{\mathbf{q}}} |n_{\mathbf{q}} - 1\rangle, \quad (6.148a)$$

$$a_{\mathbf{q}}^{\dagger} |n_{\mathbf{q}}\rangle = \sqrt{n_{\mathbf{q}} + 1} |n_{\mathbf{q}} + 1\rangle. \quad (6.148b)$$

Then the matrix element is given by

$$\begin{aligned} \langle \mathbf{k}', n_{\mathbf{q}'} | H_{\text{el}} | \mathbf{k}, n_{\mathbf{q}} \rangle &= D_{\text{ac}} \sum_{\mathbf{q}} \sqrt{\frac{\hbar}{2MN\omega_{\mathbf{q}}}} (i\mathbf{e}_{\mathbf{q}} \cdot \mathbf{q}) \sqrt{n_{\mathbf{q}} + \frac{1}{2} \mp \frac{1}{2}} \\ &\quad \times \frac{1}{L^3} \int d^3\mathbf{r} u_{\mathbf{k}'}^*(\mathbf{r}) u_{\mathbf{k}}(\mathbf{r}) e^{i(\mathbf{k} - \mathbf{k}' \pm \mathbf{q})\cdot\mathbf{r}}. \end{aligned} \quad (6.149)$$

Now, we replace the integral over the whole of space with a summation of unit cell integrations, and we obtain

$$\begin{aligned} &\langle \mathbf{k}', n_{\mathbf{q}'} | H_{\text{el}} | \mathbf{k}, n_{\mathbf{q}} \rangle \\ &= D_{\text{ac}} \sum_{\mathbf{q}} \sqrt{\frac{\hbar}{2MN\omega_{\mathbf{q}}}} (i\mathbf{e}_{\mathbf{q}} \cdot \mathbf{q}) \sqrt{n_{\mathbf{q}} + \frac{1}{2} \pm \frac{1}{2}} I(\mathbf{k}, \mathbf{k}') \delta_{\mathbf{k}', \mathbf{k} \pm \mathbf{q}}, \end{aligned} \quad (6.150)$$

$$I(\mathbf{k}, \mathbf{k}') = \frac{1}{\Omega} \int_{\Omega} d^3\mathbf{r} u_{\mathbf{k}'}^*(\mathbf{r}) u_{\mathbf{k}}(\mathbf{r}), \quad (6.151)$$

where the summation with respect to  $\mathbf{q}$  reduces to  $\delta_{\mathbf{k}', \mathbf{k} \pm \mathbf{q}}$ , where we take into account the normal process only.

Here we will show a more convenient method to derive the above result by using the Fourier transform. When we insert the Fourier transform of  $H_{\text{el}}$  into (6.140) we obtain the following result directly. Defining

$$H_{\text{el}}(\mathbf{r}) = \sum_{\mathbf{q}} [C(\mathbf{q}) a_{\mathbf{q}} e^{i\mathbf{q}\cdot\mathbf{r}} + C^{\dagger}(\mathbf{q}) a_{\mathbf{q}}^{\dagger} e^{-i\mathbf{q}\cdot\mathbf{r}}], \quad (6.152)$$

$$C(\mathbf{q}) = D_{\text{ac}} \sqrt{\frac{\hbar}{2MN\omega_{\mathbf{q}}}} (i\mathbf{e}_{\mathbf{q}} \cdot \mathbf{q}), \quad (6.153)$$

we obtain the following relation easily:

$$H_{\text{el}}(\mathbf{q}) = \frac{1}{L^3} \int d^3\mathbf{r} H_{\text{el}}(\mathbf{r}) e^{-i\mathbf{q}\cdot\mathbf{r}}$$

$$\begin{aligned}
&= \sum_{\mathbf{q}'} \left[ C(\mathbf{q}') a_{\mathbf{q}'} \frac{1}{L^3} \int d^3 \mathbf{r} e^{i(\mathbf{q}' - \mathbf{q}) \cdot \mathbf{r}} + C^\dagger(\mathbf{q}') a_{\mathbf{q}'}^\dagger \frac{1}{L^3} \int d^3 \mathbf{r} e^{-i(\mathbf{q}' + \mathbf{q}) \cdot \mathbf{r}} \right] \\
&= \sum_{\mathbf{q}'} \left[ C(\mathbf{q}') a_{\mathbf{q}'} \delta_{\mathbf{q}', \mathbf{q}} + C^\dagger(\mathbf{q}') a_{\mathbf{q}'}^\dagger \delta_{\mathbf{q}', -\mathbf{q}} \right] \\
&= C(\mathbf{q}) a_{\mathbf{q}} + C^\dagger(-\mathbf{q}) a_{-\mathbf{q}}^\dagger.
\end{aligned} \tag{6.154}$$

Inserting this relation into (6.140) we obtain

$$\begin{aligned}
&\langle n_{\mathbf{q}'} | H_{\text{el}}(\mathbf{k}' - \mathbf{k}) | n_{\mathbf{q}} \rangle \\
&= \langle n_{\mathbf{q}'} | C(\mathbf{k}' - \mathbf{k}) a_{\mathbf{k}' - \mathbf{k}} + C^\dagger(\mathbf{k} - \mathbf{k}') a_{\mathbf{k} - \mathbf{k}'}^\dagger | n_{\mathbf{q}} \rangle \\
&= \begin{cases} C(\mathbf{q}) \sqrt{n_{\mathbf{q}}} & (\mathbf{k}' = \mathbf{k} + \mathbf{q}; \text{absorption}), \\ C^\dagger(\mathbf{q}) \sqrt{n_{\mathbf{q}} + 1} & (\mathbf{k}' = \mathbf{k} - \mathbf{q}; \text{emission}). \end{cases}
\end{aligned} \tag{6.155}$$

### 6.3.3 Ionized Impurity Scattering

(a) *Brooks–Herring Formula.* The Coulomb potential due to an  $i$ th point charge  $ze$  is expressed in real space as

$$V_i(r) = \frac{ze^2}{4\pi\epsilon r}, \tag{6.156}$$

where the dielectric constant  $\epsilon$  is used to take account of the electronic polarization and ionic polarization due to lattice vibrations. The contribution of valence electrons to the dielectric constant will be discussed later. The Fourier transform of the potential gives  $V_i(\mathbf{q})$  and the calculation of scattering rate (relaxation time) is straightforward:

$$V_i(\mathbf{q}) = \frac{ze^2}{L^3 \epsilon} \frac{1}{q^2}. \tag{6.157}$$

Although the derivation of the above equation is simple, we consider the screened Coulomb potential for the purpose of calculation and generality and assume a distribution of ionic charge  $N_I$  given by

$$\rho_I(\mathbf{r}) = ze \sum_{i=1}^{N_I} \delta(\mathbf{r} - \mathbf{r}_i), \tag{6.158}$$

where  $\mathbf{r}_i$  is the position vector of an ionized impurity. The Coulomb potential of the ionized impurities is expressed as

$$V(\mathbf{r}) = \sum_{i=1}^{N_I} V_i(\mathbf{r} - \mathbf{r}_i). \quad (6.159)$$

We will discuss in Sect. 6.3.10 how to take account of the screening, and here we use the result. The screened Coulomb potential  $V(\mathbf{r} - \mathbf{r}_i)$  is given by

$$V(\mathbf{r} - \mathbf{r}_i) = \frac{ze^2}{4\pi\epsilon} \frac{e^{-q_s|\mathbf{r}-\mathbf{r}_i|}}{|\mathbf{r} - \mathbf{r}_i|}, \quad (6.160)$$

where  $q_s$  is the inverse of the Debye screening length  $\lambda_s$  and is given by

$$q_s = \frac{1}{\lambda_s} = \left( \frac{nze^2}{\epsilon k_B T} \right)^{1/2}. \quad (6.161)$$

In the above equation  $n$  is the density of electrons in the conduction band. The Fourier transform of the Coulomb potential is calculated as follows. For simplicity a single point charge is assumed and we define the potential as

$$V(r) = C \frac{e^{-q_s r}}{r}. \quad (6.162)$$

The Fourier transform is therefore given by

$$\begin{aligned} V(q) &= \frac{1}{L^3} \int V(r) e^{-i\mathbf{q} \cdot \mathbf{r}} d^3\mathbf{r} \\ &= \frac{1}{L^3} \frac{4\pi C}{q^2 + q_s^2}. \end{aligned} \quad (6.163)$$

From this relation we have

$$V(\mathbf{q}) = V(q) = \frac{ze^2}{\epsilon L^3} \frac{1}{q^2 + q_s^2}. \quad (6.164)$$

When we put  $q_s = 0$  in the above equation, (6.157) is obtained.

From these results the Coulomb potential of ionized impurities  $N_I$  is Fourier transformed to give

$$V(\mathbf{q}) = \sum_{i=1}^{N_I} \frac{ze^2}{\epsilon L^3} \frac{1}{q^2 + q_s^2} e^{-i\mathbf{q} \cdot \mathbf{r}_i}. \quad (6.165)$$

Relaxation time is calculated by inserting (6.165) into (6.137) or (6.140) and the squared matrix element of ionized impurity scattering is given by

$$\begin{aligned} |V(\mathbf{k}' - \mathbf{k})|^2 &= \left( \frac{ze^2}{\epsilon L^3} \right)^2 \frac{1}{(|\mathbf{k}' - \mathbf{k}|^2 + q_s^2)^2} \sum_{i=1}^{N_I} \sum_{j=1}^{N_I} \delta_{ij} \\ &= N_I \left( \frac{ze^2}{\epsilon L^3} \right)^2 \frac{1}{(|\mathbf{k}' - \mathbf{k}|^2 + q_s^2)^2}. \end{aligned} \quad (6.166)$$

Therefore, we obtain the relaxation time by inserting this into (6.137), resulting in

$$\frac{1}{\tau_I(\mathbf{k})} = \frac{L^3}{(2\pi)^2 \hbar} \int d^3 \mathbf{k}' N_I \left( \frac{ze^2}{\epsilon L^3} \right)^2 \frac{1}{(|\mathbf{k}' - \mathbf{k}|^2 + q_s^2)^2} (1 - \cos \theta) \times \delta[\mathcal{E}_{\mathbf{k}'} - \mathcal{E}_{\mathbf{k}}] . \quad (6.167)$$

When we define the impurity density by  $n_I = N_I/L^3$ , the relaxation time reduces to

$$\frac{1}{\tau_I(\mathbf{k})} = \frac{n_I}{(2\pi)^2 \hbar} \int d^3 \mathbf{k}' \left( \frac{ze^2}{\epsilon} \right)^2 \frac{1}{(|\mathbf{k}' - \mathbf{k}|^2 + q_s^2)^2} (1 - \cos \theta) \delta[\mathcal{E}_{\mathbf{k}'} - \mathcal{E}_{\mathbf{k}}] . \quad (6.168)$$

The electron energy is defined by

$$\mathcal{E}_{\mathbf{k}} \equiv \mathcal{E}(\mathbf{k}) = \frac{\hbar^2 k^2}{2m^*} \quad (6.169)$$

and the integral is carried out in spherical  $\mathbf{k}$  space:

$$d^3 \mathbf{k}' = 2\pi k'^2 dk' \sin \theta d\theta , \quad (6.170)$$

where  $\theta$  is the angle between  $\mathbf{k}$  and  $\mathbf{k}'$ , and  $|\mathbf{k}' - \mathbf{k}| = 2k \sin(\theta/2)$ . Using the  $\delta$ -function we find

$$\begin{aligned} \int k'^2 dk' \delta \left[ \frac{\hbar^2 k'^2}{2m^*} - \frac{\hbar^2 k^2}{2m^*} \right] &= \frac{2m^*}{\hbar^2} \int k'^2 dk' \delta[ (|\mathbf{k}'| - |\mathbf{k}|)(|\mathbf{k}'| + |\mathbf{k}|)] \\ &= \frac{2m^*}{\hbar^2} \int \frac{k'^2}{2|\mathbf{k}|} dk' \delta[ (|\mathbf{k}'| - |\mathbf{k}|)] \\ &= \frac{m^*}{\hbar^2} |\mathbf{k}| , \end{aligned} \quad (6.171)$$

and (6.168) reduces to

$$\frac{1}{\tau_I(\mathbf{k})} = \frac{n_I m^* k}{2\pi \hbar^3} \left( \frac{ze^2}{\epsilon} \right)^2 I(\mathbf{k}) , \quad (6.172)$$

where

$$\begin{aligned} I(\mathbf{k}) &= \int_0^\pi \left[ \frac{1}{\{2k \sin^2(\theta/2)\}^2 + q_s^2} \right]^2 (1 - \cos \theta) \sin \theta d\theta \\ &= \frac{1}{4k^4} \left\{ \log[1 + (2k\lambda_s)^2] - \frac{(2k\lambda_s)^2}{1 + (2k\lambda_s)^2} \right\} . \end{aligned} \quad (6.173)$$

Therefore, the relaxation time for ionized impurity scattering is given by the following equation:

$$\frac{1}{\tau_1(\mathcal{E})} = \frac{z^2 e^4 n_I}{16\pi\epsilon^2 \sqrt{2m^*}} \mathcal{E}^{-3/2} \left[ \log \left( 1 + \frac{8m^* \lambda_s^2 \mathcal{E}}{\hbar^2} \right) - \frac{8m^* \lambda_s^2 \mathcal{E} / \hbar^2}{1 + (8m^* \lambda_s^2 \mathcal{E} / \hbar^2)} \right]. \quad (6.174)$$

The above equation is called the Brooks–Herring formula.

Here we will show that the Fourier transform of a 2-dimensional Coulomb potential is given by

$$V(q) = \frac{e^2}{2\epsilon L^2} \frac{1}{q}. \quad (6.175)$$

The Fourier transform of a 2-dimensional potential  $V(r)$  is defined by

$$V(q) = \frac{1}{L^2} \int V(r) e^{i\mathbf{q} \cdot \mathbf{r}} D^2 \mathbf{r}. \quad (6.176)$$

When we express the 2-dimensional potential as

$$V(r) = \frac{e^2}{4\pi\epsilon r}, \quad (6.177)$$

then we have

$$V(q) = \frac{e^2}{4\pi\epsilon L^2} \int_0^\infty dr \int_0^{2\pi} d\phi e^{iqr \cos \phi} = \frac{e^2}{2\epsilon L^2 q} \int_0^\infty d(qr) J_0(qr), \quad (6.178)$$

where  $J_0(x)$  is the Bessel function of order zero and

$$\int_0^\infty J_0(x) dx = 1. \quad (6.179)$$

Therefore, the Fourier transform of the 2-dimensional potential is given by

$$V(q) = \frac{e^2}{2\epsilon L^2} \frac{1}{q}. \quad (6.180)$$

(b) *Conwell–Weisskopf Formula.* Conwell and Weisskopf [6.15] have derived the relaxation time for ionized impurity scattering by adopting Rutherford scattering. Let us define the scattering cross-section of a single impurity by  $A$ . In a semiconductor with impurity density  $n_I$  and with electron velocity  $v = (\partial\mathcal{E}/\partial k)/\hbar$ , the relaxation time (collision time) of an electron  $\tau_I$  is given by  $1/\tau_I = n_I v A$ . The differential cross-section  $\sigma(\theta, \phi)$  is defined by the probability of scattering into a small solid angle  $d\Omega = \sin\theta d\theta d\phi$ , and in the case of isotropic scattering the cross-section is given by  $A = \int \sigma(\theta, \phi) \sin\theta d\theta d\phi$ . Assuming elastic scattering and defining the angle  $\theta$  between the electron wave vectors  $\mathbf{k}$  and  $\mathbf{k}'$ , we obtain



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